

Bonus Part

1. Run Multiple Simulations

Currently only one experiment is run. The problem is that bandit results are stochastic, so one run is noisy. An improvement could be to run multiple independent simulations (e.g., 100 runs). And then report statistical measures: Mean cumulative reward, standard deviation, confidence intervals, and bandit selection counts. This could be a big statistical upgrade.

2. Improve Epsilon Decay Strategy

A minor improvement would be to change the epsilon decay from $1/t$ to c/t to encourage more exploration early in the experiment before gradually exploiting the optimal bandit. As a result, the algorithm is less likely to converge on and exploit a suboptimal bandit.

Example from my runs:

[E_Greedy] Arm Selection Counts:

- Arm 0 (mean=1): 1 times
- Arm 1 (mean=2): 3 times
- Arm 2 (mean=3): 19994 times
- Arm 3 (mean=4): 2 times

[Thompson] Arm Selection Counts:

- Arm 0 (mean=1): 1 times
- Arm 1 (mean=2): 1 times
- Arm 2 (mean=3): 1114 times
- Arm 3 (mean=4): 18884 times

3. Store More Informative Data

Current CSV contains only Bandit, Reward and Algorithm. Adding other components such, as timestep, cumulative reward and regret would enable deeper analysis later.